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Original Report

Can emergent treatments result in more severe errors?: An analysis of a large institutional near-miss incident reporting database

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Abstract

Purpose: Emergent radiation treatments may be subject to more errors because of the compressed time frame. Few data exist on the magnitude of this problem or how to guide safety improvement interventions. The purpose of this study is to examine patterns of near-miss events in emergent treatments using a large institutional incident reporting system.

Methods and materials: Events in the incident reporting database from February 2012 to October 2013 were reviewed prospectively by a multidisciplinary team to identify emergent treatments. Reports were scored for potential near-miss risk index (NMRI) on a 0 to 4 scale. Workflow steps of where events originated and were detected were analyzed. Events were categorized by use of the causal factor system from the Radiation Oncology Incident Learning System. Mann-Whitney U tests were used to compare mean NMRI score, and Fisher exact tests were performed to compare the proportion of high-risk events between emergent and nonemergent treatments and between emergent treatments on weekdays and weekends or holidays.

Results: Over the study period, approximately 1600 patients were treated, 190 of them emergently. Seventy-one incident reports were submitted for 55 unique patients. Fewer events were reported for emergent treatments than for nonemergent treatments (0.37 events per new treatment vs 0.86; $P < .01$). Mean risk index for emergent reports was 1.90 versus 1.48 for nonemergent reports ($P < .01$). Rate of NMRI 4 was 10% for emergent treatments versus 4% for nonemergent treatments ($P < .01$). Emergent treatments started on a weekend or holiday had a higher proportion of critical near-miss events than emergent treatments started during the week (37% vs 7.9%, $P = .034$).

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Conclusions: In this study, fewer near-miss incidents were reported per treatment course for emergent treatments. This may be attributable to reporting bias. More importantly, when emergent near misses occur, they are of greater severity.

Introduction

Lifesaving emergency medical care occurs at thousands of emergency departments (EDs) nationwide every day, but the same interventions can also potentially harm patients, whether by misdiagnosis,[1] medication errors,[2] medication overdoses,[3] or charting and ordering on the wrong patient.[4] A study conducted in Swedish EDs found that human error and the ED environment, driven by lack of standard procedures, lack of resources, and high workload, were major factors that contributed to errors.[5] Safety concerns relating to emergency medical care are now being recognized, and efforts have been made to establish a culture of safety that is inclusive of nonphysician groups, such as the emergency medical services group[6] and pharmacists.[7] Incident reporting and learning are being implemented in EDs with good participation of health care providers.[8] Vulnerable areas are being specifically targeted for intervention, including standardization of communication during shift handoffs[9,10] and prominent displaying of a patient's room number on electronic medical records to prevent writing orders for and charting on the wrong patient.[11]

The same issues and challenges that arise in an ED may also apply to the process of providing emergent treatment in radiation oncology. Oncological emergencies, such as spinal cord compression, intractable pain, neurologic decline, intractable bleeding, and superior vena cava syndrome, often require emergent radiation treatment. Oncological emergencies can occur on weekends or holidays, during which there are limitations to the availability of treatment staff, the potential for alternative workflows, and compressed time frames. Although numerous studies have been published on patient safety initiatives in radiation oncology,[12-14] and there is active participation in the new Radiation Oncology Incident Learning System (RO-ILS), there is a dearth of studies pertaining to safety initiatives and emergent treatments.

The purpose of this study was to examine the patterns of near-miss events in emergent treatments using our large institutional near-miss incident learning and reporting system.

Methods and materials

In February 2012, an incident reporting and learning system (ILS) was implemented at our institution. This system was based on recommendations from the American Association of Physicists in Medicine (AAPM) Work Group on Prevention of Errors in Radiation Oncology[15] and has been described previously.[16,19] The time period selected for this study ranged from February 2012 to October 2013. All incidents in the system were reported voluntarily. Each week, a multidisciplinary team reviewed reported incidents and assigned near-miss severity scores by consensus but was not blinded to the details of the event, specifically as to whether the incident was associated with an emergent or nonemergent treatment. For learning and process improvement purposes, the database does not specifically exclude events that reached the patient. Our institution has a hospital-wide reporting system for events that reached the patient, which is used in conjunction with the ILS.

Treatments associated with the reports were identified as starting on a weekday or weekend/holiday. Within the ILS data system, each event is prospectively categorized according to keywords assigned by the review committee. Events that had been keyword tagged as related to emergent/weekend/holiday treatment were identified in the ILS. The electronic medical record (MOSAIQ, Elekta, Stockholm, Sweden) was also queried to identify emergent/weekend treatments by running a custom report of patients who had orders that defined emergent treatment and their emergency type. The data were cross-referenced with the ILS database to identify additional reports related to emergent treatments. All charts of the identified patients were reviewed to ensure that the intent of treatment was emergent.

Each report was assigned a near-miss risk index (NMRI), ranging from 0 to 4 (none, mild, moderate, severe, critical). The details of the NMRI have been described previously.[19] Briefly, NMRI 0 events do not impact patient safety or quality of treatment; NMRI 1 events may enhance downstream risks or cause patient inconvenience; NMRI 2 events enhance the risk of critical errors downstream; NMRI 3 events have potential clinical impact, with limited barriers for problem prevention; and NMRI 4 events have potentially critical clinical impact, with extremely limited barriers for prevention.

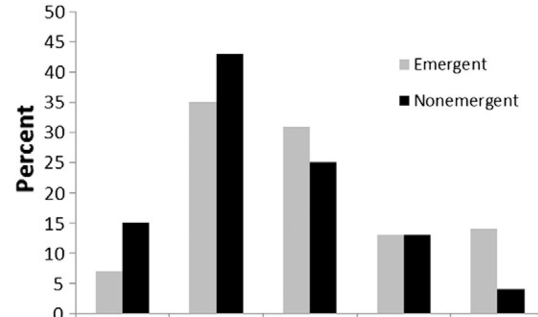
The workflow step in which the reported event originated and was discovered was also recorded, with the steps based on the AAPM consensus guideline.[15] Additionally, the proportion of NMRI 4 events was chosen to be evaluated for comparison between weekday and weekend/holiday emergent treatments because these were the most severe incidents, and we hypothesized they were most likely to be reported and captured in the ILS. Mann-Whitney *U* tests were used to compare mean NMRI score, and Fisher exact statistical tests were performed to compare proportions of NMRI.

Emergent treatment near-miss events were analyzed for root causes with the "causal factors" categorization used in the RO-ILS system, which matches that of the AAPM consensus report on incident learning.[15] There are 7 major root cause

Table 1 Emergent versus nonemergent treatments

	Emergent treatments	Nonemergent treatments	*P* value
Total, n	190	1600	
Reported events, n	71	1383	
Near-miss events per treatment	0.37	0.86	<.01
Mean NMRI	1.9	1.48	.009
NMRI score, n (%)			
0	5 (7)	211 (15)	
1	25 (35)	596 (43)	
2	22 (31)	341 (25)	
3	9 (13)	177 (13)	
4	10 (14)	58 (4)	.002
NMRI 4 for all treatments	5% (10/190)	4% (58/1410)	.42

NMRI, Near-Miss Risk Index.



	0	1	2	3	4
Emergent	7%	35%	31%	13%	14%
Nonemergent	15%	43%	25%	13%	4%

Figure 1 Distribution of NMRI (Near-Miss Risk Index) score for emergent and nonemergent treatments.

categories, consisting of (1) organizational management, (2) technical, (3) human behavior involving staff, (4) patient-focused circumstances, (5) external factors (beyond facility control), (6) procedural issues, and (7) other. Each event was reviewed and categorized on the basis of what was thought to be the major contributing root cause based on the details of the event, with the acknowledgement that an event can have more than 1 root cause.

This study was approved by the institutional review board.

Results

A query of the electronic medical record from February 2012 to October 2013 identified 190 treatments for oncological emergencies. Indications for emergent treatment were intractable pain (42%), progressive neurologic symptoms (24%, with 35% of those treatments involving spinal cord compression), rapid progression of tumor (15%), intractable bleeding (6%), and superior vena cava syndrome (3%).

In the study time frame, there were 1454 total near-miss events reported to the ILS for approximately 1600 treatments. There were 1383 nonemergent treatment near-miss events reported, with a ratio of approximately 0.86 reports per new treatment course for nonemergent patients. There were 71 near-miss events for 190 total new emergent treatments during the study time frame, with a ratio of 0.37 reports per new emergent treatment course, which was significantly lower than nonemergent treatment ($P < .01$). Details of emergent near-miss events versus nonemergent near-miss events are outlined in Table 1.

More emergent events (71) were reported than the number of treatment courses reported (55) because 8 treatments (15%) had multiple events reported: 3 patient treatments with 4 event reports, 2 patient treatments with 3 event reports, and 3 patient treatments with 2 events. For nonemergent treatments, 723 treatments (52%) had multiple near-miss events reported, with the breakdown as follows: 1 treatment had 11 events, 2 had 9 events, 10 had 8 events, 8 had 7 events, 16 had 6 events, 44 had 5 events, 78 had 4 events, 171 had 3 events, and 393 had 2 events.

Near-miss risk index

The mean NMRI score of emergent treatments was 1.90, which was significantly higher than the mean NMRI score of 1.48 for nonemergent treatments ($P = .009$). The distribution of NMRI scores for emergent versus nonemergent treatments was 7% (5/71) versus 15% (211/1383) for events with an NMRI of 0, 35% (25/71) versus 43% (596/1383) for an NMRI of 1, 31% (22/71) versus 25% (341/1383) for an NMRI of 2, 13% (9/71) versus 13% (177/1383) for an NMRI of 3, and 14% (10/71) versus 4% (58/1383) for an NMRI of 4 (Fig 1). Emergent treatment events reported had a significantly higher proportion of NMRI 4 (14%, 10/71) than nonemergent treatments with events reported (6%, 78/1383; $P = .002$). The rate of NMRI 4 for all emergent treatments was 5% (10/190), and for all nonemergent treatments, it was 4% (58/1410), which was not statistically significantly different ($P = .43$).

When we considered all the NMSI 4 emergent near-miss events, 37% were for treatments initiated on a weekend or holiday, whereas 8% were for treatments initiated during the week ($P = .003$) (Fig 2). This is despite the fact that treatments initiated on weekends or holidays make up only 3% (54/1600) of treatments at our institution. The NMRI 4 incidents involved a shift in vertebral levels treated (the vertebra level intended to be treated was still in the field); an isocenter that was changed after being marked, which resulted in a discrepancy of documented shifts; and a custom electron cut out calculated with monitoring units for a similar electron circular cut out.

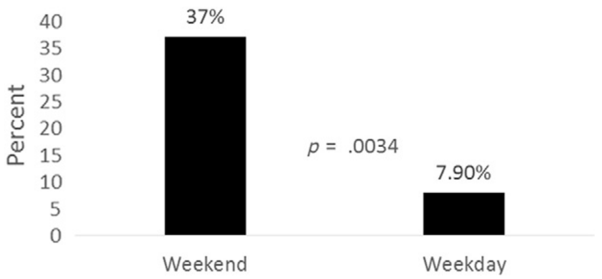


Figure 2 Proportion of NMRI (Near-Miss Risk Index) score 4 for emergent treatment started on a weekend versus that started on a weekday.

Event origination and detection

For emergent treatments, more than half of all near-miss events (53%) originated at treatment planning (Fig 3). This was significantly higher than for nonemergent treatments, for which only 33% of events originated at this step ($P < .001$) (Fig 3). Distribution of origination points was not significantly different for all other workflow steps (7%-10%). When we examined a more granular level the steps within treatment planning, nearly a quarter of all events (23%) originated at the substep of "plan information transfer to radiation oncology information system." The other prominent subcategory for treatment planning event was "dose distribution calculation" (14%).

The workflow step where most (42%) of the events associated with emergent treatment were discovered was "pretreatment plan check," compared with 25% for nonemergent treatments (P = .0012). The distribution of origination points was not significantly different for all other workflow steps. Prominent subcategories within the pretreatment plan check were "physics plan review," where 30% of all total emergent near-miss events were detected, and "therapist chart check," where 11% of all total events were detected. "Treatment delivery" accounted for 25% of discovery points, and "on treatment review" accounted for 18% of discovery points (Fig 4).

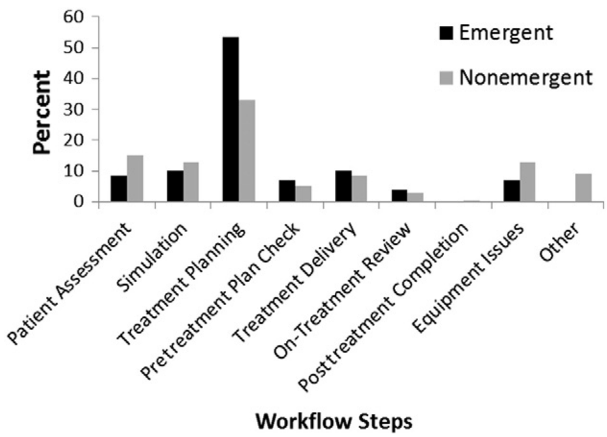


Figure 3 Origination points of emergent near-miss events.

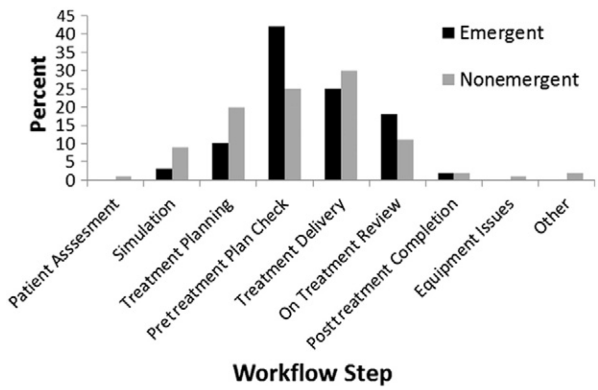


Figure 4 Discovery points of emergent near-miss events.

Root cause factors

Categorization of each event showed that 3 major root cause categories were applicable to 97% of all events: organizational management (category 1), human behavior (category 3), and procedural issues (category 6). In each of these 3 categories, specific subcategories dominated. In "organizational management," 66% of events categorized (20/30) pertained to the subcategory of "communication," which constituted 28% of all events. In "human behavior involving staff," 82% of events (9/11) pertained to the subcategory of "poor judgment (failure to carry out quality control on patient due to time limitation)." In "procedural issues," 64% of events (18/28) pertained to the subcategory of "failure to detect a developing problem," which constituted 25% of all events. Data are shown in Table 2.

Discussion

This study used institutional safety initiatives and a near-miss ILS to investigate errors associated with emergent radiation treatments. Approximately 12% of patients (190/1600) treated during the 20-month study period were treated for emergent indications. Twenty-nine percent of those patients (55/190) had a near-miss incident reported, which constituted 5% of all incident reports (71/1454) over the study period.

The results of this study indicate that fewer near-miss events were reported per treatment course for emergent treatments despite the fact that emergent treatments are often characterized by compacted timelines, limited staffing, and altered treatment pathway processes. This difference may be affected by reporting bias and differences in the nature of emergent and conventional treatments, as discussed below. However, when near misses occur, they appear to be of greater severity. The mean NMRI score for emergent treatments was higher (1.9 vs 1.4, P = .009), and the proportion of NMSI 4 events was higher in emergent

Table 2 Root cause factors as categorized by the casual factor categorization used in RO-ILS

Causal factor	Percentage of emergent ILS reports with this root cause factor	Examples of events
1. Organizational management d. Communication	28	• Lack of information regarding bolus in the treatment field • No communication to dosimetry of CBCT needs, and patient was planned on a machine without CBCT capacity • No communication between SIM group and dosimetry personnel regarding immobilization with an arm board that was not indexed because it was used for patient girth and not for setup
6. Procedural issues a. Failure to detect a developing problem iii. Loss of attention	25	• DRR was associated with the wrong field • Normalization point left at isocenter instead of the calculation point used for dose calculation before plan transfer • Forgetting to print out the prescription dose line
3. Human behavior involving staff c. Poor judgment (eg, failure to carry out quality control on a patient due to time limitations)	13	• Not documenting a shift from isocenter • Not completing couch removal in the treatment planning system

CBCT, cone beam computed tomography; DRR, digitally reconstructed radiography; ILS, incident learning system; RO-ILS, Radiation Oncology Incident Learning System; SIM, simulation.

treatments (14% vs 5%, $P = .002$). Emergent treatments that began on weekends or holidays were found to have a greater proportion of high-severity NMRI 4 scores than emergent treatment begun on weekdays.

Examination of these high-severity incidents helped spur a change in policy and implementation of new procedures in our clinic. For example, 1 weekend treatment incident involved a shift in level of vertebral bodies on a patient who was transferred from a satellite clinic to continue emergent treatment without undergoing another simulation. This altered the patient's setup, because different immobilization devices were used. This event was reported to the hospital-wide patient safety system, and a root cause analysis was performed. From this, a new transfer policy was implemented, which required resimulation for all patients transferred from other clinics.

One limitation of this study is that not all near-miss events related to emergent treatment may have been reported to the ILS. For example, 71 near-miss events for 55 patients were reported for 190 emergent treatments, which resulted in a ratio of 0.37 events per treatment, which was statistically significantly lower than reporting rates for nonemergent treatments, with a ratio of 0.86 events per treatment. It is possible that staff members were less likely to report noncritical near-miss events because of the shortened timeline associated with weekend and emergent treatments, which would represent reporter bias. Another reason may be that emergent treatments have fewer handoffs because of reduced clinic staff and streamlined workflows. Anecdotally, communication and handoffs between staff and treatments contribute to a large portion of lower NMRI events in the ILS. Additionally, emergent treatments may be associated with fewer complex techniques than other treatments (eg, stereotactic body radiation therapy, breath-hold control, 4-dimensional computed tomography simulations), which would represent fewer chances for near misses to occur in the emergent pathway.

The difference in high-severity events (NMRI 4) is particularly interesting, because we hypothesize that these events are less subject to reporting bias. These events are of such critical importance that they are very likely to be reported in the system used here. The higher proportion of NMRI 4 events in emergent treatments may therefore be a real effect.

Regarding workflow categorization, the majority of incidents (53%) from emergent radiation treatments originated at the treatment planning stage, which was a significantly higher proportion than for nonemergent treatments (33%). Transferring data has been recognized by the AAPM as an area that needs extra attention, with the publication of a rapid communication from Task Group 201.[17] One potential means to prevent events from emerging from treatment planning is to use a checklist or automated plan checking software. In response to data from the ILS, our department has since developed an automated program for detecting common errors, written in-house in the Pinnacle scripting language, although it was not used during this study period. This could potentially significantly reduce incidents occurring at the "plan information transfer" stage, where 23% of all origination points occurred.

When we examined detection points for incidents, we discovered that 42% of events were detected at "pretreatment review and verification," with 2 subcategories, "physics plan review" and "therapy chart check," accounting for 41% of events (30% and 11% of discovery points, respectively). A previous study using an alternate methodology suggested that the physics chart check is approximately 63% effective for detecting near miss events.[18] Some departments do not require that a physics check be performed before the patient's first treatment, especially for treatment courses planned on a weekend. It is within the American College of Radiology–American Society for Radiation Oncology guidelines to perform this task within the first 3 treatments.

Although checklists are gaining prominence in radiation oncology and are the subject of the forthcoming AAPM Task Group 230, emergent treatments may be an area in which they are even more useful. When checklists and timeouts have been implemented in radiation oncology departments, they have reduced actual patient errors, including those involving treatment of the wrong site and deliverance of the wrong dose.[13]

Every department has its own process for emergent treatments, especially those initiated during the weekend, and our findings have informed the development of improvements at our institution. As more institutions implement departmental ILS initiatives, the information can be used to study patterns of events for emergent treatments, possibly in conjunction with the national RO-ILS system.

In conclusion, this study suggests that fewer near-miss reports per treatment course are associated with emergent radiation treatments than with standard treatments; however, when near misses occur, they are of greater severity. Common points of event origination and detection, such as during treatment planning and at physics chart check and therapist plan review, were identified. These results have contributed to policy and procedural changes in our department and may be useful to other departments that seek to evaluate their own emergent treatment processes.

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